

Cultural Screenplay and Visual Adaptation Studio

Build a working application that recreates a short screenplay inside an exact Indian culture and dialect, then produces a consistent visual production pack.

LEVEL	AI Engineer / Senior AI Engineer
DEADLINE	48 hours from assignment receipt
INPUT	A supplied 3-5 page screenplay with 3-5 scenes
REQUIRED PRODUCT	Working web MVP, adapted screenplay and visual pack
CORE SIGNAL	Cultural depth, structured extraction and continuity

This challenge rewards judgment, evidence, verification and safe autonomy. A polished demo without a trustworthy decision process will not score well.

1. Challenge overview

Build a simple web application that accepts a short screenplay and re-creates it inside a selected Indian culture. The user chooses an exact culture or dialect, the region, whether the setting is rural or urban, and the writing script for the output.

The application must adapt both the screenplay and its production world. The result should feel as if the story belongs naturally to the selected culture. A word-for-word translation with different clothes or names is not enough.

What success looks like

If dialogue is hidden, the generated characters, costumes, locations, props and scene visuals should still communicate the selected culture. If visuals are hidden, the dialogue, relationships, humour and behaviour should still feel culturally specific.

2. Input, user choice and output

Input	User chooses	System produces
A 3-5 page screenplay pasted into the app or uploaded as TXT, DOCX or PDF.	One or more exact cultures or dialects; region; rural or urban setting; and output writing script.	A separate adapted screenplay and visual production pack for every selected culture.

Examples of acceptable precision include **Majhi, Malwai, Bangru, Marwari** or **Mewari**. Labels such as 'Punjabi culture', 'Rajasthani look' or 'North Indian village' are too broad unless the user explicitly requests that level.

3. What must be adapted

Dimension	The adaptation should consider
Verbal	Dialect, idioms, kinship terms, honorifics, humour, rhythm, politeness, formality and code-switching.
Non-verbal	Gesture, posture, eye contact, touch, silence, greeting, seating, eating and personal space.
Characters	Age, class, work, family role, personality, confidence, restraint, affection, anger, relationships and emotional arc.
Visual world	Architecture, interiors, wardrobe, fabric, colour, jewellery, grooming, props, food, transport and landscape.
Story world	Motivations, conflict, stakes, community influence, rituals, humour, resolution, sound and atmosphere.
Cultural precision	A specific region and dialect, with clear handling of uncertainty and no accidental mixing of unrelated traditions.

Preserve the story

The cultural world may change, but the central dramatic purpose, character relationships and emotional arc of the source should remain recognizable unless the system clearly explains why a deeper change is required.

4. Required product flow

Step	User experience	Required system behaviour
1. Upload	Paste or upload a screenplay.	Read the document, preserve scene order and report extraction problems.
2. Select	Choose culture, exact dialect, region, setting and output script.	Keep every selected culture as an independent adaptation.
3. Review	Inspect extracted scenes, characters, costumes and adaptation plan.	Allow correction, merging, renaming and approval before images are generated.
4. Generate	Create screenplay adaptation and visual assets.	Use approved canonical records and maintain continuity across every output.
5. Export	Download the adapted screenplay and production pack.	Export structured data, readable reports and generated images together.

5. Scene-by-scene extraction

Before adaptation or image generation, extract the screenplay into structured records. The review screen should make incorrect entities and continuity errors easy to find.

Record	Minimum information
Scene	Scene number, INT/EXT, location, sub-location, time, day or date, weather, mood, short summary and dramatic purpose.
Character	Canonical ID, name, aliases, age, role, relationships, personality, dialect, emotional state and entrance or exit.
Production elements	Set, costume, grooming, jewellery, props, food, vehicles, animals, extras, rituals, gestures and sound or music.
State and continuity	What each character wears, carries, knows, loses, gains or changes before and after a scene, including injuries and prop transfers.

A structured scene record may resemble:

```
{
  "scene_id": "SC03",
  "location_id": "LOC_HOME_COURTYARD",
  "time": "early morning",
  "characters": ["CHAR_AMAR", "CHAR_BEBE"],
  "costume_ids": ["COST_AMAR_01", "COST_BEBE_01"],
  "props_in": ["PROP_LETTER"],
  "props_out": [],
  "emotional_change": "Amar moves from avoidance to admission",
  "continuity_warnings": []
}
```

6. Canonical identity and continuity

Your system must prevent duplicate or inconsistent production elements.

- Merge aliases and repeated mentions into one canonical character record.
- Create one canonical record for each recurring location, prop and costume; reuse its ID across scenes.

- Do not generate multiple character images for the same approved appearance.
- Keep face, body, apparent age, grooming and key identity features consistent across images.
- Keep a costume until the story gives a believable reason for changing it.
- Track props, knowledge, injuries, relationships and emotional state between scenes.
- Flag contradictions and show every affected scene before generation.

Example

If 'Amar', 'the elder son' and 'A. Singh' refer to the same person, the application should create one character ID. If that character carries a letter out of Scene 2, the letter cannot disappear in Scene 3 without an explanation.

7. Cultural adaptation requirements

The application should first create an adaptation plan, allow the user to review it, and then rewrite the screenplay. The plan should explain important cultural decisions rather than silently changing the story.

- Adapt names, work, family roles, community influence and social expectations where needed.
- Rewrite dialogue for dialect, rhythm, humour, kinship and respect, not only vocabulary.
- Adapt behaviour and gesture as well as spoken text.
- Choose setting, objects, food, transport and sound that belong to the exact region and time period.
- Preserve story logic. A cultural change must not create a new plot hole or reverse a character's motivation.
- Record uncertain cultural choices and allow the user or reviewer to correct them.
- Keep two selected cultures separate. Do not allow prompts, memory or assets from one adaptation to leak into another.

8. Visual generation is mandatory

Generate visual assets only after the user approves the extracted characters, cultural adaptation plan and costume plan. Generated visuals should be culturally recognizable even when dialogue is hidden.

Visual pack	Required output
Character bible	For every important character: canonical ID, role, age, physical description, grooming, personality cues and one consistent full-body reference image.
Costume bible	Generate each unique costume variant once, including full-body view, garments, fabrics, colours, footwear, jewellery, headwear, grooming and scenes where it appears.
Scene look	Generate one culturally specific keyframe for every scene showing location, architecture, character blocking, approved costumes, props, time, lighting and atmosphere.

The visual pipeline should use approved canonical references when generating later images. Include the model, prompt or structured generation specification needed to reproduce each asset.

9. Minimum application screens

Screen	Minimum capability
Upload and culture selection	Accept input, choose exact culture or dialect, setting and output script.
Scene extraction	Display scenes, characters, production elements and continuity state in an editable table.
Character and costume approval	Merge duplicates, approve canonical records and review cultural plans before generation.
Adapted screenplay comparison	Show the source and adapted versions with important changes and explanations.
Visual gallery	Browse images by character, costume and scene, with linked canonical IDs.
Continuity and export	Show contradictions, affected scenes and downloadable output packages.

10. Agentic engineering expectations

This is an AI engineering assignment, not only a prompt-writing exercise. Your design should show:

- A clear sequence of extraction, normalization, planning, approval, adaptation, generation and verification.
- Structured memory for canonical characters, costumes, locations, props and scene state.
- Tools or checks that can reject duplicate entities, broken continuity, missing scenes or inconsistent visuals.
- Human approval gates before expensive or hard-to-reverse visual generation.
- A way to retry one failed asset without rebuilding the entire project.
- Traceability from adapted dialogue and images back to the source scene and approved cultural plan.
- Clear handling of model failure, uncertain cultural knowledge and incomplete source documents.

11. Mandatory 48-hour acceptance test

Test input	Target	Pass condition
A supplied 3-5 page neutral screenplay containing 3-5 scenes and 3-6 characters.	One assigned exact culture and dialect. A second independent adaptation is optional bonus work.	Accurate scene extraction; no duplicate characters or costumes; consistent continuity; native-feeling dialogue and action; recognizable character, costume and scene visuals; and no unintended cultural mixing.

Priority rule

A compact and working MVP is preferred over a broad unfinished system. If time is limited, complete one culture end to end before adding a second culture or advanced UI features.

12. Required deliverables

- Working web application and complete source repository.
- README with setup steps, system requirements and a sample environment file.
- Sample adapted screenplay, structured scene breakdown and continuity report.
- Generated character bible, costume bible and scene visual pack.
- Automated tests, known limitations and AI-usage log.

- A 3-5 minute recorded demonstration plus a live technical explanation.

```

submission/
README.md
ARCHITECTURE.md
AI_USAGE.md
KNOWN_LIMITATIONS.md
app/
tests/
sample_output/
adapted_screenplay.pdf
scene_breakdown.json
continuity_report.pdf
character_bible/
costume_bible/
scene_keyframes/
demo_video.mp4

```

13. Evaluation

Area	Weight	What we will look for
Extraction and continuity	25%	Accurate structure, canonical entities, state tracking and useful contradiction detection.
Cultural adaptation	20%	Specific, coherent verbal and non-verbal adaptation without generic decoration or cultural mixing.
Visual consistency	20%	Recognizable culture and stable character, costume, location and prop identity across images.
Working product and UX	15%	The required flow works and review or approval steps are easy to understand.
No duplication	10%	Aliases, costumes, locations and reusable assets are correctly normalized and reused.
Code, tests and explanation	10%	Clear implementation, useful tests, reproducible setup and honest explanation of limitations.

What will score poorly

A word-for-word translation with decorative clothing; generic cultural labels; duplicate character identities; costumes that change without reason; visuals generated before approval; beautiful images with broken continuity; or a large interface whose core flow does not work.

14. Demonstration and discussion

During the live explanation, be prepared to demonstrate:

- One character from source mention to canonical record, adapted identity and consistent visual output.
- One costume or prop as it moves across multiple scenes.
- One cultural adaptation decision that required more than translation.
- One contradiction detected before visual generation.
- How a user corrects an extraction or cultural choice and regenerates only affected outputs.
- How AI assistants were used, what they got wrong and how their output was verified.